

Unit-3

Relational query Language





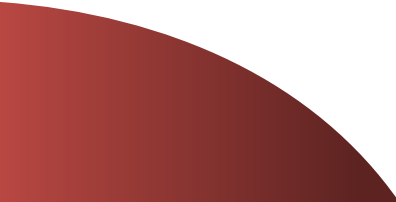
Outline

- Structure of Relational Databases
- Key
- Relational Algebra
 - Fundamental Operators and Syntax
 - Selection
 - Projection
 - Cross Product OR Cartesian Product
 - Joins
 - Set Operators
 - Division
 - Rename
 - Aggregate Functions
- Open Source and Commercial DBMS



Structure of Relational Databases

Section - 1



Structure of Relational Databases

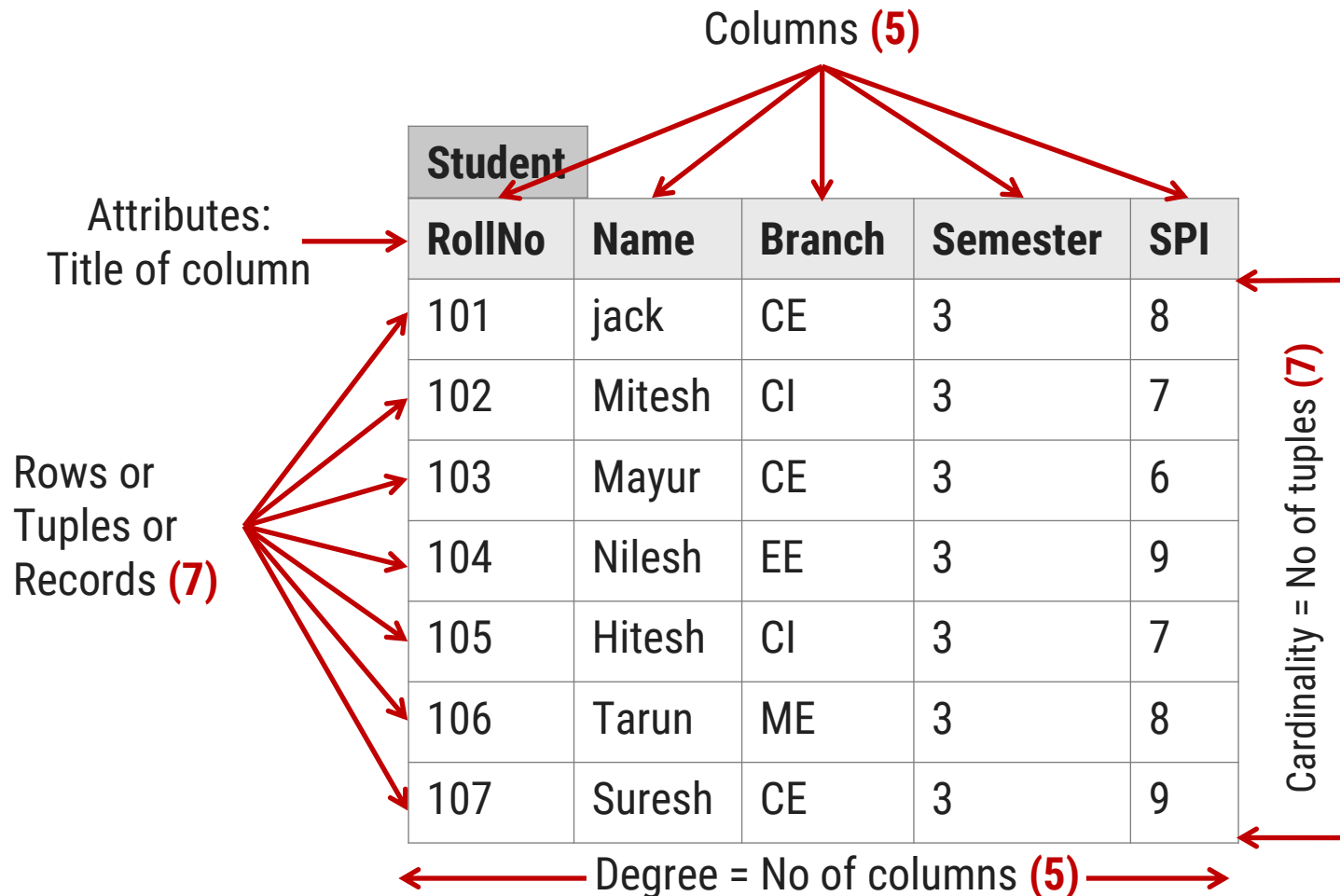


Table (Relation): A database object that holds a collection of data for a specific topic. Table consist of rows and columns.

Column (Attribute): The vertical component of a table. A column has a name and a particular data type; e.g. varchar, decimal, integer, datetime etc.

Record (Tuple): The horizontal component of a table, consisting of a sequence of values, one for each column of the table. It is also known as row.

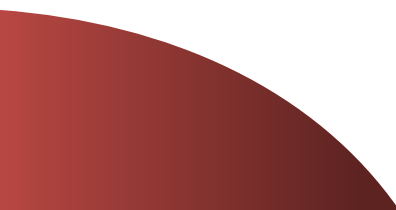
A database consists of a collection of tables (relations), each having a unique name.

Domain is a set of **all possible unique values** for a specific column. Domain of Branch attribute is (CE, CI, ME, EE)



Key

Section - 2



Super Key

- ▶ A super key is a set of one or more **attributes whose values uniquely identifies each record** within a relation (table).

Super Key EnrollNo	Super Key (RollNo, Branch, Sem)			Super Key (SPI, Name, BL)		
EnrollNo	RollNo	Branch	Sem	SPI	Name	BL
190540107001	101	CE	3	8	jack	0
190540107002	102	CE	3	7	Mitesh	1
190540106001	101	CI	3	6	Mayur	2
190540106002	102	CI	3	9	Nilesh	0
180540107001	101	CE	5	7	Hitesh	1
180540106001	101	CI	5	8	Tarun	0
180540106002	102	CI	5	9	Suresh	0

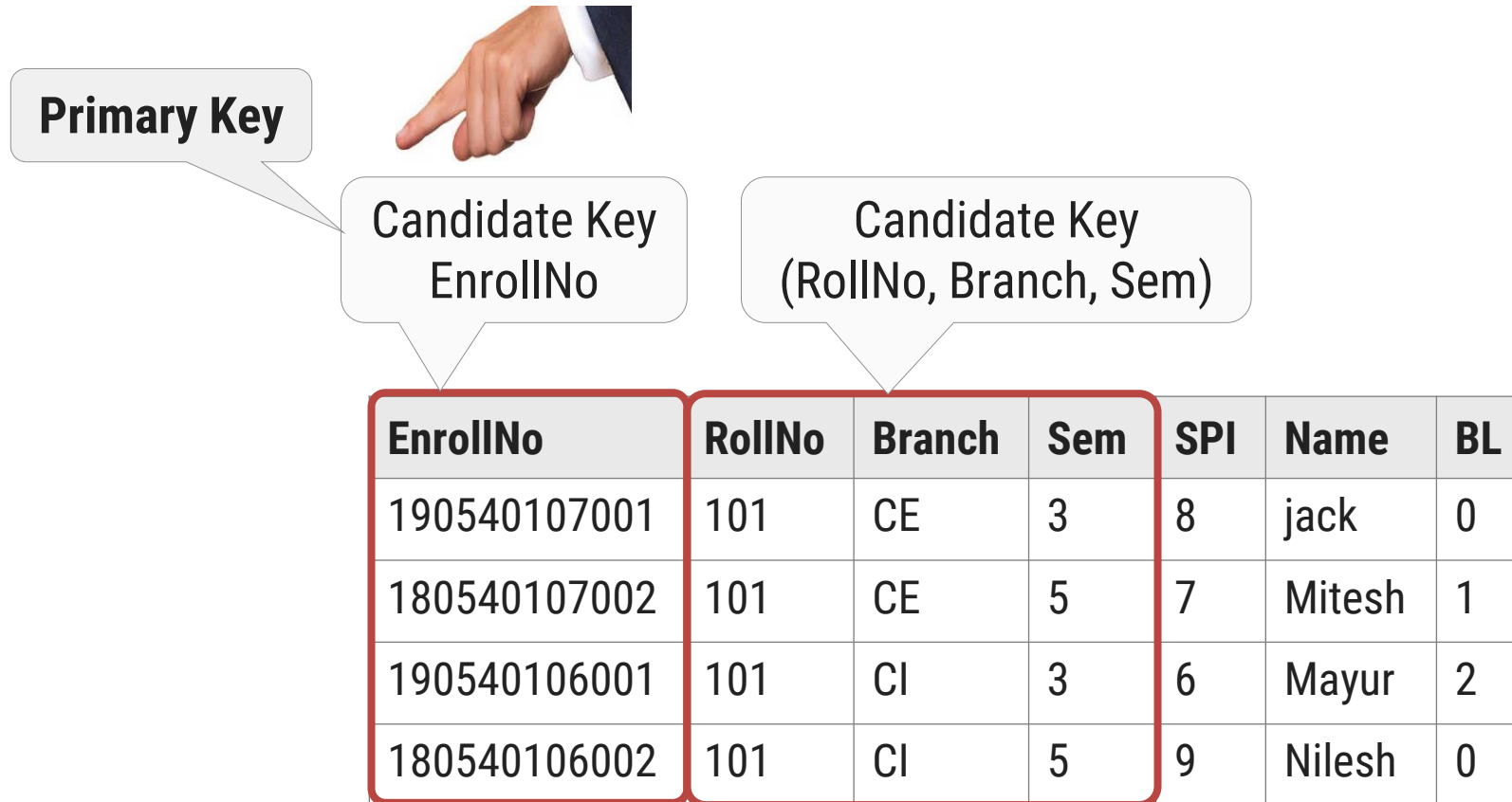
Candidate Key

- ▶ A candidate key is a **subset of a super key**.
- ▶ A candidate key is a single attribute or the least combination of attributes that uniquely identifies each record in the table.
- ▶ A candidate key is a **super key for which no proper subset is a super key**.
- ▶ **Every candidate key is a super key** but **every super key is not a candidate key**.

Candidate Key EnrollNo		Candidate Key (RollNo, Branch, Sem)				
EnrollNo	RollNo	Branch	Sem	SPI	Name	BL
190540107001	101	CE	3	8	jack	0
180540107002	101	CE	5	7	Mitesh	1
190540106001	101	CI	3	6	Mayur	2
180540106002	101	CI	5	9	Nilesh	0

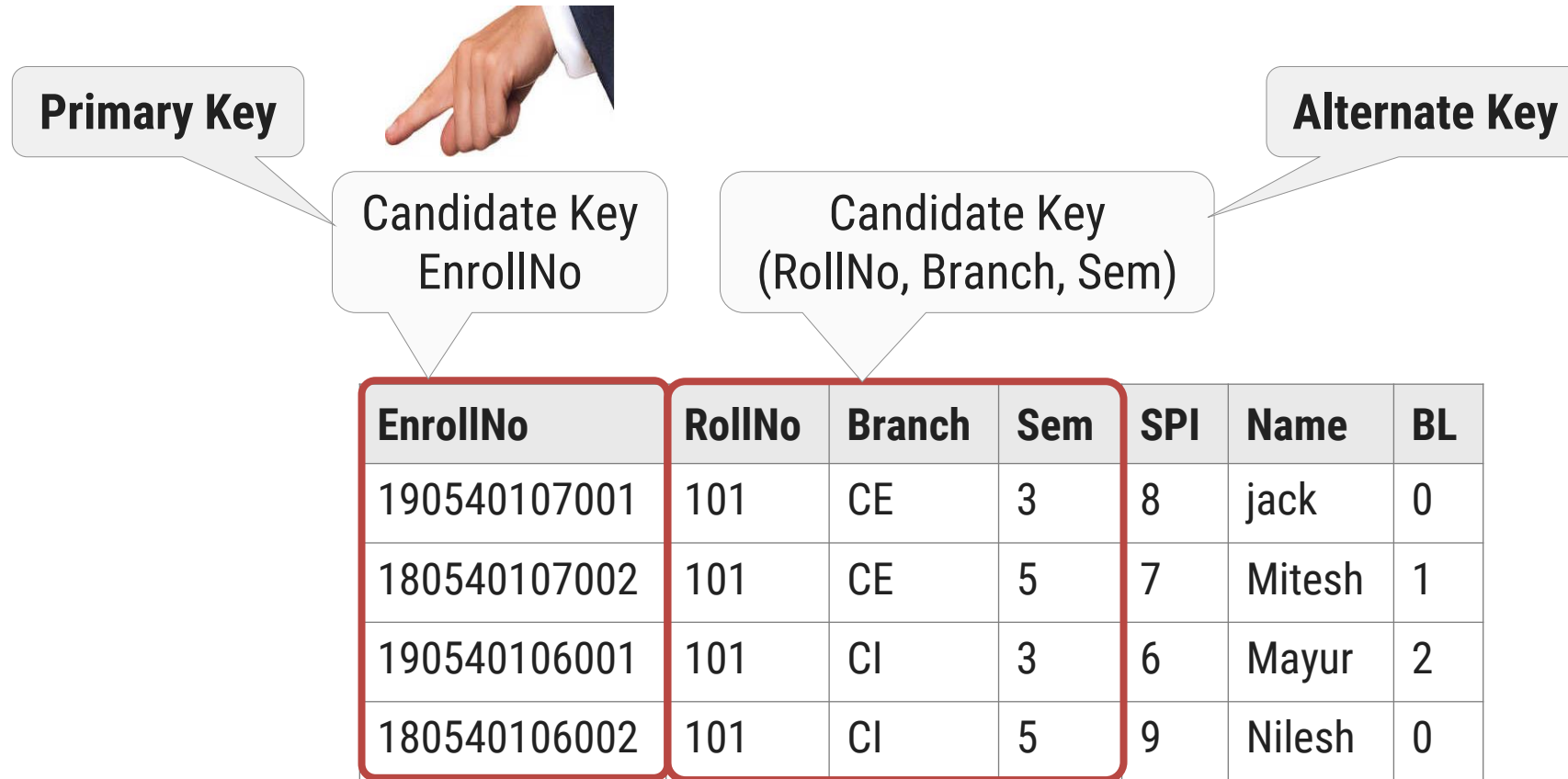
Primary Key

- ▶ A primary key is a **candidate key that is chosen by database designer** to identify tuples uniquely in a relation (table).



Alternate Key

- ▶ An alternate key is a **candidate key that is not chosen by database designer** to identify tuples uniquely in a relation.

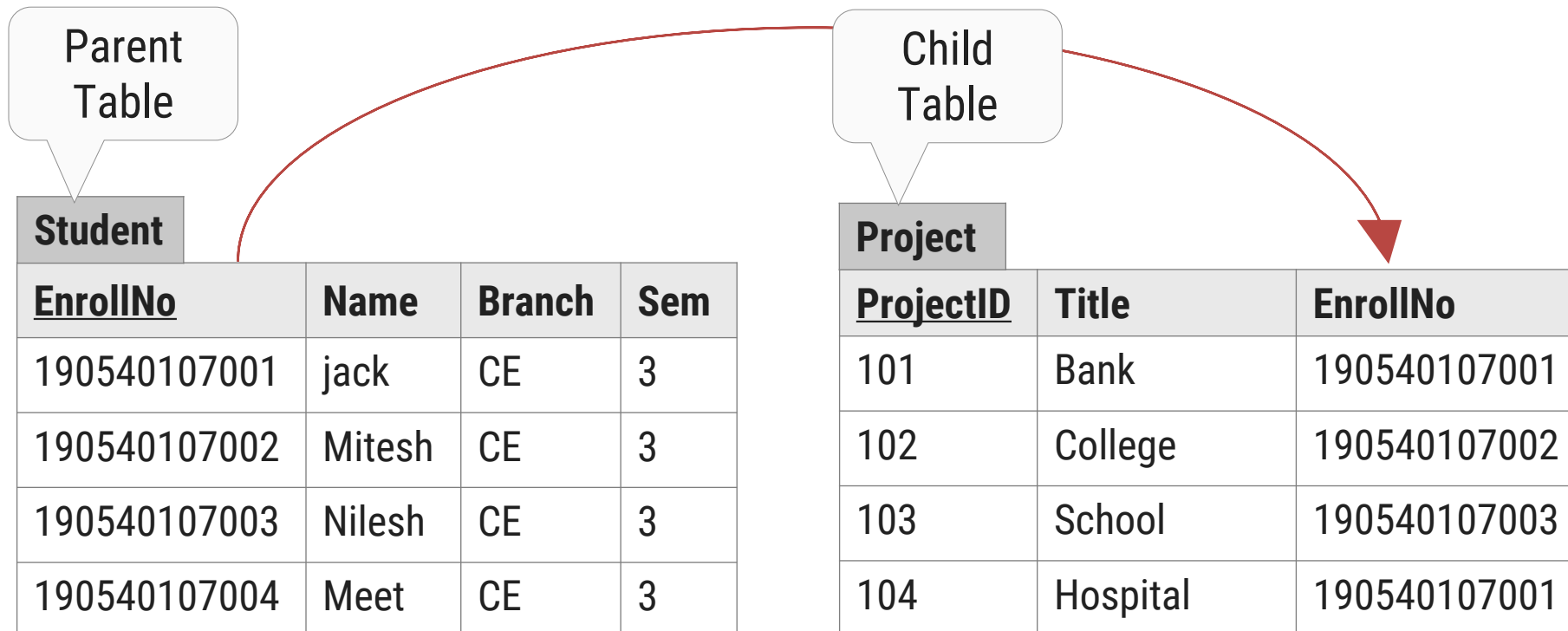


Primary Key rules

- ▶ A primary key **may have one or more attributes**.
- ▶ There is **only one primary key** in the relation (table).
- ▶ A primary key **attribute value cannot be NULL**.
- ▶ Generally, the **value of a primary key attribute does not change**.

Foreign Key

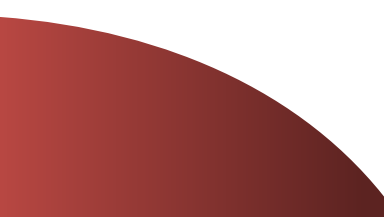
- ▶ A foreign key is **used to link two relations** (tables).
- ▶ A foreign key is an **attribute** or collection of attributes in one table that **refers to the primary key in another table**.
- ▶ A table containing the foreign key is called the child table, and the table containing the primary key is called the parent table.





Relational Algebra Operations

Section - 3



Relational Algebra Operations

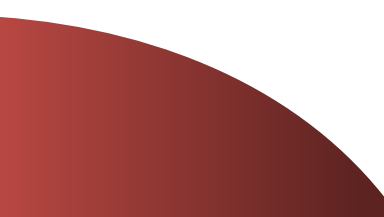
Operator	Description
Selection	Display particular rows/records/tuples from a relation
Projection	Display particular columns from a relation
Cross Product	Multiply each tuples of both relations
Joins	Combine data or records from two or more tables 1. Natural Join / Inner Join 2. Outer Join 1. Left Outer Join 2. Right Outer Join 3. Full Outer Join
Set Operators	Combine the results of two queries into a single result. 1. Union 2. Intersection 3. Minus / Set-difference
Division	Divides one relation by another
Rename	Rename a column or a table



Relational Algebra Operations

Selection Operator

Section - 3.1



Selection Operator

- ▶ Symbol: σ (Sigma)
- ▶ Notation: $\sigma_{condition}$ (Relation)
- ▶ Operation: **Selects tuples** from a relation that **satisfy a given condition**.
- ▶ Operators: =, <>, <, >, <=, >=, $\wedge$ (AND), $\vee$ (OR)

Example Display the detail of students belongs to “CE” Branch.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	9

Answer $\sigma_{Branch='CE'}$ (Student)

Output			
RollNo	Name	Branch	SPI
101	jack	CE	8
104	Meet	CE	9

Selection Operator [$\sigma_{\text{condition}}$ (Relation)]

Example Display the detail of students belongs to “CE” Branch and having SPI more than 8.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	9

Answer $\sigma_{\text{Branch}='CE' \wedge \text{SPI}>8}$ (Student)

Output			
RollNo	Name	Branch	SPI
104	Meet	CE	9

Selection Operator [$\sigma_{\text{condition}}$ (Relation)]

Example Display the detail of students belongs to either “CI” or “ME” Branch.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	9

Answer $\sigma_{\text{Branch}='CI' \vee \text{Branch}='ME'}$ (Student)

Output			
RollNo	Name	Branch	SPI
102	Mitesh	ME	9
103	Nilesh	CI	9

Selection Operator [$\sigma_{\text{condition}}$ (Relation)]

Example Display the detail of students whose SPI between 7 and 9.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	9

Answer $\sigma_{SPI > 7 \wedge SPI < 9}$ (Student)

Output			
RollNo	Name	Branch	SPI
101	jack	CE	8

Exercise

► Write down the relational algebra for the student table.

- Display the detail of students whose RollNo is less than 104.
- Display the detail of students having SPI more than 8.
- Display the detail of students belongs to “CE” Branch having SPI less than 8.
- Display the detail of students belongs to either “CE” or “ME” Branch.
- Display the detail of students whose SPI between 6 and 9.

Student			
RollNo	Name	Branch	SPI
101	Raj	CE	6
102	Meet	ME	8
103	Harsh	EE	7
104	Punit	CE	9

► Write down the relational algebra for the employee table.

- Display the detail of all employee.
- Display the detail of employee whose Salary more than 10000.
- Display the detail of employee belongs to “HR” Dept having Salary more than 20000.
- Display the detail of employee belongs to either “HR” or “Admin” Dept.
- Display the detail of employee whose Salary between 10000 and 25000 and belongs to “HR” Dept.

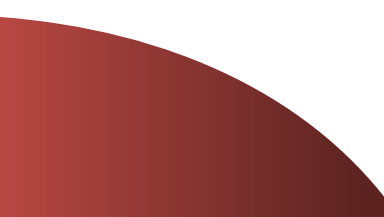
Employee			
EmpID	Name	Dept	Salary
101	Nilesh	Sales	10000
102	Mayur	HR	25000
103	Hardik	HR	15000
104	Ajay	Admin	20000



Relational Algebra Operations

Projection Operator

Section - 3.2



Projection Operator

- ▶ Symbol: Π (Pi)
- ▶ Notation: Π *attribute set* (Relation)
- ▶ Operation: **Selects specified attributes** of a relation.
- ▶ It **removes duplicate tuples** (records) from the result.

Example Display **RollNo**, **Name** and **Branch** of all students.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	9

Answer Π *RollNo, Name, Branch* (Student)

Output		
RollNo	Name	Branch
101	jack	CE
102	Mitesh	ME
103	Nilesh	CI
104	Meet	CE

Exercise

► Write down the relational algebra for the student table.

- ➔ Display RollNo, Name and SPI of all students.
- ➔ Display Name and SPI of all students.
- ➔ Display the Name of all students.
- ➔ Display the Name of all branches.

Student			
RollNo	Name	Branch	SPI
101	Raj	CE	6
102	Meet	ME	8
103	Harsh	EE	7
104	Punit	CE	9

► Write down the relational algebra for the employee table.

- ➔ Display EmpID with Name of all employee.
- ➔ Display Name and Salary of all employee.
- ➔ Display the Name of all employee.
- ➔ Display the Name of all departments.

Employee			
EmpID	Name	Dept	Salary
101	Nilesh	Sales	10000
102	Mayur	HR	25000
103	Hardik	HR	15000
104	Ajay	Admin	20000

Combined Projection & Selection Operation

Example Display RollNo, Name & Branch of “ME” Branch students.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	7

Step-1 $\sigma_{\text{Branch}='ME'}$ (Student)

Output-1			
RollNo	Name	Branch	SPI
102	Mitesh	ME	9

Answer $\Pi_{\text{RollNo, Name, Branch}} (\sigma_{\text{Branch}='ME'} (\text{Student}))$

Output-2		
RollNo	Name	Branch
102	Mitesh	ME

Combined Projection & Selection Operation

Example Display **Name, Branch and SPI** of students whose **SPI is more than 8**.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	7

Step-1 $\sigma_{SPI > 8}$ (Student)

Output-1			
RollNo	Name	Branch	SPI
102	Mitesh	ME	9
103	Nilesh	CI	9

Answer $\Pi_{Name, Branch, SPI} (\sigma_{SPI > 8} (\text{Student}))$

Output-2		
Name	Branch	SPI
Mitesh	ME	9
Nilesh	CI	9

Combined Projection & Selection Operation

Example Display **Name, Branch and SPI** of students who belongs to “CE” Branch and SPI is more than 7.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	7

Step-1 $\sigma_{\text{Branch}='CE' \wedge \text{SPI}>7}$ (Student)

Output-1			
RollNo	Name	Branch	SPI
101	jack	CE	8

Answer $\Pi_{\text{Name, Branch, SPI}} (\sigma_{\text{Branch}='CE' \wedge \text{SPI}>7} (\text{Student}))$

Output-2		
Name	Branch	SPI
jack	CE	8

Combined Projection & Selection Operation

Example Display **Name** of students along with their **Branch** who belong to either “ME” Branch or “CI” Branch.

Student			
RollNo	Name	Branch	SPI
101	jack	CE	8
102	Mitesh	ME	9
103	Nilesh	CI	9
104	Meet	CE	7

Step-1 $\sigma_{\text{Branch}='ME' \vee \text{Branch}='CI'} (\text{Student})$

Output-1			
RollNo	Name	Branch	SPI
102	Mitesh	ME	9
103	Nilesh	CI	9

Answer $\Pi_{\text{Name}, \text{Branch}} (\sigma_{\text{Branch}='ME' \vee \text{Branch}='CI'} (\text{Student}))$

Output-2	
Name	Branch
Mitesh	ME
Nilesh	CI

Exercise

► Write down the relational algebra for the student table.

- ➔ Display Rollno, Name and SPI of all students belongs to “CE” Branch.
- ➔ List the Name of students with their Branch whose SPI is more than 8 and belongs to “CE” Branch.
- ➔ List the Name of students along with their Branch and SPI who belongs to either “CE” or “ME” Branch and having SPI more than 8.
- ➔ Display the Name of students with their Branch name whose SPI between 7 and 9.

Student			
RollNo	Name	Branch	SPI
101	Raj	CE	6
102	Meet	ME	8
103	Harsh	EE	7
104	Punit	CE	9

► Write down the relational algebra for the employee table.

- ➔ Display the Name of employee belong to “HR” Dept and having salary more than 20000.
- ➔ Display the Name of all “Admin” and “HR” Dept’s employee.
- ➔ List the Name of employee with their Salary who belongs to “HR” or “Admin” Dept having salary more than 15000.
- ➔ Display the Name of employee along with their Dept name whose salary between 15000 and 30000.

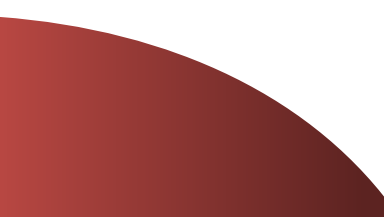
Employee			
EmpID	Name	Dept	Salary
101	Nilesh	Sales	10000
102	Mayur	HR	25000
103	Hardik	HR	15000
104	Ajay	Admin	20000



Relational Algebra Operations

Cartesian Product / Cross Product

Section - 3.3



Cartesian Product / Cross Product

- ▶ Symbol: $\times$ (Cross)
- ▶ Notation: *Relation-1 (R1) $\times$ Relation-2 (R2)* **OR** *Algebra-1 $\times$ Algebra-2*
- ▶ Operation: It will **multiply each tuples** of Relation-1 to each tuples of Relation-2.
 - ➔ Attributes of Resultant Relation = Attributes of R1 + Attributes of R2
 - ➔ Tuples of Resultant Relation = Tuples of R1 * Tuples of R2

Example Perform Cross Product between Student and Result.

Answer (Student) $\times$ (Result)

Student			Result	
RNo	Name	Branch	RNo	SPI
101	jack	CE	101	8
102	Mitesh	ME	102	9

Output				
Student.RNo	Name	Branch	Result.RNo	SPI
101	jack	CE	101	8
101	jack	CE	102	9
102	Mitesh	ME	101	8
102	Mitesh	ME	102	9

If both relations have some attribute with the same name, it can be distinguished by combining **relation-name.attribute-name**.

Cartesian Product / Cross Product Example

Example Perform Cross Product between Student and Result.

Consider only **selected attributes**

- Student – RNo, Name and Branch
- Result – RNo, SPI and BL

Student			
RNo	Name	Branch	Sem
101	jack	CE	3
102	Mitesh	ME	5

Result			
RNo	SPI	BL	Rank
101	8	1	2
103	9	0	1

Answer $\Pi_{RNo, Name, Branch} (Student) \times \Pi_{RNo, SPI, BL} (Result)$

Output					
Student.RNo	Name	Branch	Result.RNo	SPI	BL
101	jack	CE	101	8	1
101	jack	CE	103	9	0
102	Mitesh	ME	101	8	1
102	Mitesh	ME	103	9	0

Cartesian Product / Cross Product Example

Example Perform Cross Product between Student and Result.

Consider only **selected tuples**

- Student – Branch='CE' and Sem=3
- Result – SPI>7 and BL<1

Student			
RNo	Name	Branch	Sem
101	jack	CE	3
102	Mitesh	ME	5
103	Om	CE	3
104	Dhara	CE	5

Result			
RNo	SPI	BL	Rank
101	8	1	2
103	9	0	1
105	7	2	3

Answer $\sigma_{\text{Branch}='CE' \wedge \text{Sem}=3}(\text{Student}) \times \sigma_{\text{SPI}>7 \wedge \text{BL}<1}(\text{Result})$

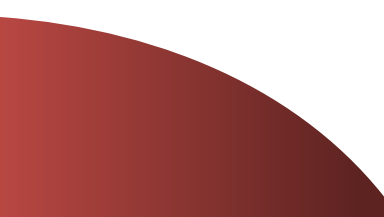
Output							
Student.RNo	Name	Branch	Sem	Result.RNo	SPI	BL	Rank
101	jack	CE	3	103	9	0	1
103	OM	CE	3	103	9	0	1



Relational Algebra Operations

Natural Join / Inner Join

Section - 3.4

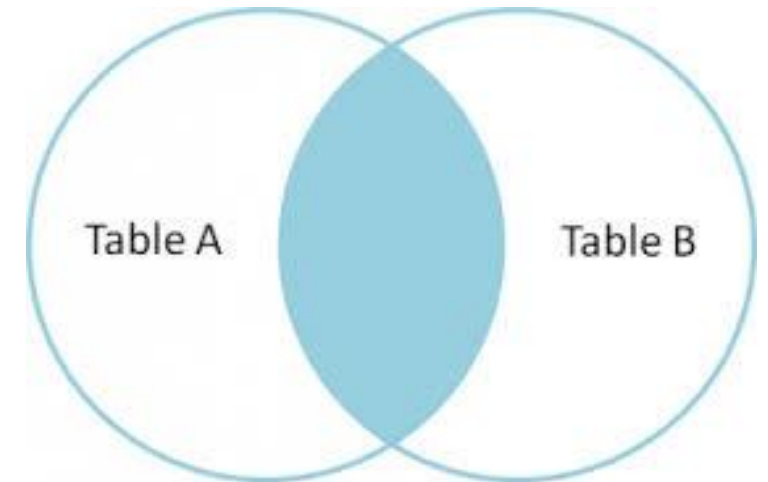


Natural Join / Inner Join

- ▶ Symbol: $\bowtie$
- ▶ Notation: $Relation-1 (R1) \bowtie Relation-2 (R2)$ OR $Algebra-1 \bowtie Algebra-2$
- ▶ Operation: Natural join will **retrieve consistent data** from multiple relations.
 - ➔ It **combines records** from different relations that **satisfy a given condition**.

Steps performed in Natural Join

Steps	Description
Step – 1	It performs Cartesian Product
Step – 2	Then it deletes inconsistent tuples
Step – 3	Then it removes an attribute from duplicate attributes



Natural Join / Inner Join Example

Example Perform Natural Join between Student and Result.

Student		
RNo	Name	Branch
101	jack	CE
102	Mitesh	ME

Result	
RNo	SPI
101	8
103	9

Answer (Student) ⋈ (Result)

Output			
RNo	Name	Branch	SPI
101	jack	CE	8

To perform a Natural Join there must be **one common attribute (column)** between two relations.

Steps performed in Natural Join

Step:1 Perform Cross Product

Student.RNo	Name	Branch	Result.RNo	SPI
101	jack	CE	101	8
101	jack	CE	103	9
102	Mitesh	ME	101	8
102	Mitesh	ME	103	9

Step:2 Removes inconsistent tuples

Student.RNo	Name	Branch	Result.RNo	SPI
101	jack	CE	101	8

Step:3 Removes an attribute from duplicate

RNo	Name	Branch	SPI
101	jack	CE	8

Natural Join / Inner Join Example

Example Perform Natural Join between Branch and Faculty.

Branch		
<u>BID</u>	BName	HOD
1	CE	Shah
2	ME	Patel

Faculty		
<u>FID</u>	FName	BID
101	Raj	1
103	Meet	2

Answer (Branch) ⋈ (Faculty)

Output				
BID	Bname	HOD	FID	FName
1	CE	Shah	101	Raj
2	ME	Patel	103	Meet

To perform a Natural Join there must be **one common attribute (column)** between two relations.

Write down relational algebra for the following tables/relations

► Relations

- Student (Rno, Sname, Address, City, Mobile)
- Department (Did, Dname)
- Academic (Rno, Did, SPI, CPI, Backlog)
- Guide (Rno, PName, Fid)
- Faculty (Fid, Fname, Subject, Did, Salary)

Example List the **name of students** with their **department name** and **SPI** of all student belong to “CE” department.

Answer $\Pi_{Sname, Dname, SPI} (\sigma_{Dname='CE'} (Student \bowtie (Department \bowtie Academic)))$

Example Display the **name of students** with their **project name** whose **guide** is “A. J. Shah”.

Answer $\Pi_{Sname, Pname} (\sigma_{Fname='A.J.Shah'} (Student \bowtie (Guide \bowtie Faculty)))$

Exercise: Write down relational algebra for the following tables/relations

► Relations

- ➔ Student (Rno, Sname, Address, City, Mobile)
- ➔ Department (Did, Dname)
- ➔ Academic (Rno, Did, SPI, CPI, Backlog)
- ➔ Guide (Rno, PName, Fid)
- ➔ Faculty (Fid, Fname, Subject, Did, Salary)

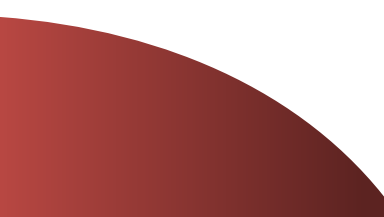
- List the **name of students** with their **department name** having **backlog 0**.
- List the **name of faculties** with their **department name** and **salary** having **salary more than 25000** and **belongs to “CE” department**.
- List the **name of all faculties** of “CE” and “ME” department whose **salary is more than 50000**.
- Display the **students name** with their **project name** of all “CE” department’s students whose **guide is “Z.Z. Patel”**.
- Display the **name of faculties** with their **department name** who belongs to “CE” department and **tough “CPU” subject** having **salary more than 25000**.
- List the **name of students** with their **department name** doing **project “Hackathon”** under **guide “I. I. Shah”**.



Relational Algebra Operations

Outer Join

Section - 3.5



Outer Join

- In **natural join** some records are missing, if we **want that missing records** than we have to **use outer join**.

Three types of Outer Join

Sr.	Outer Join	Symbol
1	Left Outer Join	
2	Right Outer Join	
3	Full Outer Join	

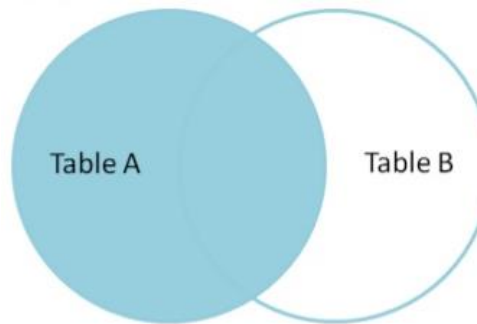
To perform a Outer Join there must be **one common attribute (column)** between two relations.

Left Outer Join

- ▶ Symbol: $\bowtie$
- ▶ Notation: $Relation-1 (R1) \bowtie Relation-2 (R2)$ OR $Algebra-1 \bowtie Algebra-2$
- ▶ Operation:
 - ➔ Display **all the tuples of the left relation** even through there is no matching tuple in the right relation.
 - ➔ For such kind of **tuples having no matching**, the attributes of right relation will be **padded with NULL** in resultant relation.

Example Perform Left Outer Join between Student and Result.

Student			Result	
RollNo	Name	Branch	RollNo	SPI
101	Raj	CE	101	8
102	Meet	ME	103	9



Answer $(Student) \bowtie (Result)$

Output			
RollNo	Name	Branch	SPI
101	Raj	CE	8
102	Meet	ME	NULL

Exercise What is the output of $(Result) \bowtie (Student)$.

Left Outer Join Example

Example Perform Left Outer Join between Student and Result. (Display RollNo, Name and SPI)

Student			Result		
RollNo	Name	Branch	RollNo	SPI	BL
101	Raj	CE	101	8	1
102	Meet	ME	103	9	0

Answer $\Pi_{RollNo, Name, SPI} ((Student) \bowtie (Result))$

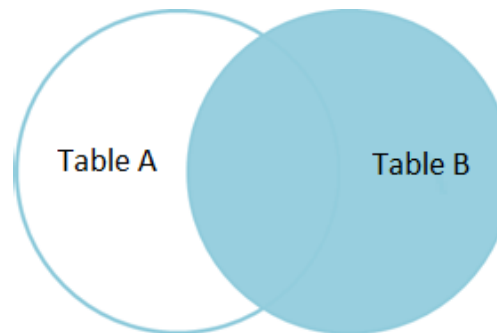
Output		
RollNo	Name	SPI
101	Raj	8
102	Meet	NULL

Right Outer Join

- ▶ Symbol: $\bowtie\!\!\!\diagup$
- ▶ Notation: $Relation-1 (R1) \bowtie\!\!\!\diagup Relation-2 (R2)$ **OR** $Algebra-1 \bowtie\!\!\!\diagup Algebra-2$
- ▶ Operation:
 - ➔ Display **all the tuples of right relation** even through there is no matching tuple in the left relation.
 - ➔ For such kind of **tuples having no matching**, the attributes of left relation will be **padding with NULL** in resultant relation.

Example Perform Right Outer Join between Student and Result.

Student			Result	
RollNo	Name	Branch	RollNo	SPI
101	Raj	CE	101	8
102	Meet	ME	103	9



Answer $(Student) \bowtie\!\!\!\diagup (Result)$

Output			
RollNo	Name	Branch	SPI
101	Raj	CE	8
103	NULL	NULL	9

Exercise What is the output of $(Result) \bowtie\!\!\!\diagup (Student)$.

Right Outer Join Example

Example Perform Right Outer Join between Student and Result. (Display RollNo, Name and SPI)

Student			Result		
RollNo	Name	Branch	RollNo	SPI	BL
101	Raj	CE	101	8	1
102	Meet	ME	103	9	0

Answer $\Pi_{RollNo, Name, SPI} ((Student) \bowtie (Result))$

Output		
RollNo	Name	SPI
101	Raj	8
103	NULL	9

Full Outer Join

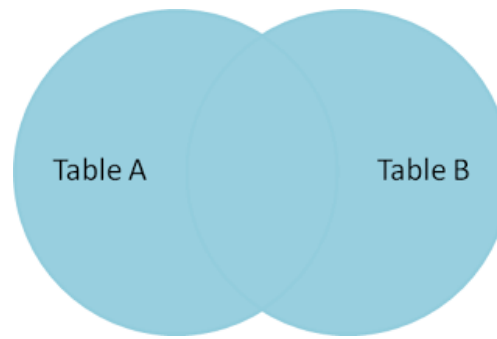
- ▶ Symbol: $\bowtie$
- ▶ Notation: $Relation-1 (R1) \bowtie Relation-2 (R2)$ OR $Algebra-1 \bowtie Algebra-2$
- ▶ Operation:
 - ➔ Display **all the tuples of both of the relations**. It also pads null values whenever required. (Left outer join + Right outer join)
 - ➔ For such kind of **tuples having no matching**, it will be **padding with NULL** in resultant relation.

Example Perform Full Outer Join between Student and Result.

Answer $(Student) \bowtie (Result)$

Student		
RollNo	Name	Branch
101	Raj	CE
102	Meet	ME

Result	
RollNo	SPI
101	8
103	9



Output			
RollNo	Name	Branch	SPI
101	Raj	CE	8
102	Meet	ME	NULL
103	NULL	NULL	9

Exercise What is the output of $(Result) \bowtie (Student)$.

Full Outer Join Example

Example Perform Full Outer Join between Student and Result. (Display RollNo, Name and SPI)

Student			Result		
RollNo	Name	Branch	RollNo	SPI	BL
101	Raj	CE	101	8	1
102	Meet	ME	103	9	0

Answer $\Pi_{RollNo, Name, SPI} ((Student) \bowtie (Result))$

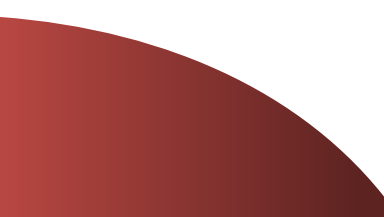
Output		
RollNo	Name	SPI
101	Raj	8
102	Meet	NULL
103	NULL	9



Relational Algebra Operations

Set Operators

Section - 3.6



Set Operators

- Set operators **combine the results of two or more queries** into a single result.

Three types of Set Operators

Sr.	Set Operator	Symbol
1	Union	U
2	Intersect / Intersection	$\cap$
3	Minus / Set difference	-

Conditions Set operators will take two or more queries as input, which must be **union-compatible**.

- Both queries should have **same (equal) number of columns**
- Corresponding **attributes should have the same data type or domain**

Conditions to perform Set Operators

Conditions-1 Both queries should have **same (equal) number of columns**.



Student			
RNo	Name	Dept	SPI
101	Raj	CE	8
102	Meet	ME	9
103	Jay	CE	9

Faculty		
Fld	Name	Dept
101	Patel	CE
102	Shah	ME
103	Dave	ME



Student		
RNo	Name	Dept
101	Raj	CE
102	Meet	ME
103	Jay	CE

Faculty		
Fld	Name	Dept
101	Patel	CE
102	Shah	ME
103	Dave	ME

Conditions-2 Corresponding **attributes should have the same data type**.



Student			
RNo	Name	Dept	SPI
101	Raj	CE	8
102	Meet	ME	9
103	Jay	CE	9

Faculty			
Fld	Name	Dept	Sub
101	Patel	CE	DS
102	Shah	ME	DBMS
103	Dave	ME	DF



Student			
RNo	Name	Dept	SPI
101	Raj	CE	8
102	Meet	ME	9
103	Jay	CE	9

Faculty			
Fld	Name	Dept	Exp
101	Patel	CE	5
102	Shah	ME	3
103	Dave	ME	4

Set Operators [Exercise]

Exercise Check whether following tables are compatible or not:

- A: (First_name(char), Last_name(char), Date_of_Birth(date))
 - B: (FName(char), LName(char), PhoneNumber(number))
 - X (Not compatible) Both tables have 3 attributes but **third attributes datatype is different.**
-
- A: (First_name(char), Last_name(char), Date_of_Birth(date))
 - B: (FName(char), LName(char), DOB(date))
 - ✓ (Compatible) Both tables have 3 attributes and of same data type.
-
- Person (PersonID, Name, Address, Hobby)
 - Professor (ProfessorID, Name, OfficeAddress, Salary)
 - (Not compatible) Both tables have 4 attributes but **forth attributes datatype is different.**
- $\Pi_{Name, Address} (Person) \quad \& \quad \Pi_{Name, OfficeAddress} (Professor)$
- (Compatible) Both tables have 2 attributes and of same data type.

Union Operator

- ▶ Symbol: $\cup$
- ▶ Notation: *Relation-1 (R1) $\cup$ Relation-2 (R2)* **OR** *Algebra-1 $\cup$ Algebra-2*
- ▶ Operation:
 - ➔ It displays all the tuples/records belonging to the first relation (left relation) or the second relation (right relation) or both.
 - ➔ It also **eliminates duplicate tuples** (tuples present in both relations appear once)

Example Perform Union between Customer and Employee.

Customer
Name
jack
Suresh
Meet

Employee
Name
Meet
Suresh
Manoj

Answer (Customer) $\cup$ (Employee)

Output
Name
Manoj
Meet
jack
Suresh

Exercise Is there any difference in the output if we swap the tables in Union operator. (Employee) $\cup$ (Customer).

Intersect/ Intersection Operator

- ▶ Symbol: $\cap$
- ▶ Notation: *Relation-1 (R1) $\cap$ Relation-2 (R2)* **OR** *Algebra-1 $\cap$ Algebra-2*
- ▶ Operation:
 - ➔ It displays all the tuples/records belonging to both relations. OR
 - ➔ It displays all the tuples/records which are common from both relations.

Example Perform Intersection between Customer and Employee.

Customer
Name
jack
Suresh
Meet

Employee
Name
Meet
Suresh
Manoj

Answer (Customer) $\cap$ (Employee)

Output
Name
Meet
Suresh

Exercise Is there any difference in the output if we swap the tables in Intersection. (Employee) $\cap$ (Customer).

Minus/ Set difference Operator

- ▶ Symbol: –
- ▶ Notation: *Relation-1 (R1) – Relation-2 (R2)* **OR** *Algebra-1 – Algebra-2*
- ▶ Operation:
 - ➔ It displays all the tuples/records belonging to the first relation (left relation) but not in the second relation (right relation).

Example Perform Set difference between Customer and Employee.

Customer
Name
jack
Suresh
Meet

Employee
Name
Meet
Suresh
Manoj

Answer (Customer) – (Employee)

Output
Name
jack

Exercise Is there any difference in the output if we swap the tables in Set difference. (Employee) – (Customer).

Union Operators Example

Example Display Name of person who are **either employee or customer**.

Customer			Employee			
ID	Name	Balance	ID	Name	Dept	Salary
1	jack	10000	2	Suresh	CE	8000
2	Suresh	20000	3	Manoj	ME	9000

Answer $\Pi_{Name}(\text{Customer}) \cup \Pi_{Name}(\text{Employee})$

Output
Name
Manoj
jack
Suresh

Intersect/ Intersection Operators Example

Example Display Name of person who are **employee as well as customer**.

Customer		
ID	Name	Balance
1	jack	10000
2	Suresh	20000

Employee			
ID	Name	Dept	Salary
2	Suresh	CE	8000
3	Manoj	ME	9000

Answer $\Pi_{Name}(\text{Customer}) \cap \Pi_{Name}(\text{Employee})$

Output	
Name	
Suresh	

Minus/ Set difference Operators Example

Example Display Name of person who are **employee but not customer**.

Customer		
ID	Name	Balance
1	jack	10000
2	Suresh	20000

Employee			
ID	Name	Dept	Salary
2	Suresh	CE	8000
3	Manoj	ME	9000

Answer $\Pi_{Name}(\text{Employee}) - \Pi_{Name}(\text{Customer})$

Output	
Name	
Manoj	

Minus/ Set difference Operators Example

Example Display Name of person who are customer but not employee.

Customer		
ID	Name	Balance
1	jack	10000
2	Suresh	20000

Employee			
ID	Name	Dept	Salary
2	Suresh	CE	8000
3	Manoj	ME	9000

Answer $\Pi_{Name}(\text{Customer}) - \Pi_{Name}(\text{Employee})$

Output	
Name	
jack	

Set Operators [Exercise]

Exercise What is the output of following relational algebra for the below mentioned tables:

Customer		
ID	Name	Balance
1	jack	10000
2	Suresh	20000

Employee			
ID	Name	Dept	Salary
2	Suresh	CE	8000
3	Manoj	ME	9000

Algebra-1 $\Pi_{ID, Name} (Customer) \cup \Pi_{ID, Name} (Employee)$

Algebra-2 $\Pi_{ID, Name, Balance} (Customer) \cup \Pi_{ID, Name, Salary} (Employee)$

Algebra-3 $\Pi_{ID, Name} (Customer) \cap \Pi_{ID, Name} (Employee)$

Algebra-4 $\Pi_{ID, Name, Balance} (Customer) \cap \Pi_{ID, Name, Salary} (Employee)$

Set Operators [Exercise]

Exercise What is the output of following relational algebra for the below mentioned tables:

Customer		
ID	Name	Balance
1	jack	10000
2	Suresh	20000

Employee			
ID	Name	Dept	Salary
2	Suresh	CE	8000
3	Manoj	ME	9000

Algebra-1 $\Pi_{ID, Name} (Customer) - \Pi_{ID, Name} (Employee)$

Algebra-2 $\Pi_{ID, Name, Balance} (Customer) - \Pi_{ID, Name, Salary} (Employee)$

Algebra-3 $\Pi_{ID, Name} (Employee) - \Pi_{ID, Name} (Customer)$

Algebra-4 $\Pi_{ID, Name, Balance} (Employee) - \Pi_{ID, Name, Salary} (Customer)$

EXCEPT

- ▶ **EXCEPT** : contains all the rows that are returned by the first SELECT operation, and not returned by the second SELECT operation.

```
SELECT column-1, column-2 .....  
FROM table 1  
WHERE....
```

EXCEPT

```
SELECT column-1, column-2 .....  
FROM table 2  
WHERE....
```

Example : Table 1 containing Employee Details

ID	Name	Age	City
1	Suresh	24	Delhi
2	Ramesh	23	pune
3	Kashish	34	Agra

Table 2 containing details of employees who are provided bonus

Bonus_ID	Employee_ID	Bonus (in RS.)
43	1	20,000
45	3	30,000

```
SELECT ID, Name, Bonus
FROM
table1
LEFT JOIN
table2
ON table1.ID = table2.Employee_ID

EXCEPT

SELECT ID, Name, Bonus
FROM
table1
RIGHT JOIN
table2
ON table1.ID = table2.Employee_ID;
```

Result :

ID	Name	Bonus
2	Ramesh	Null

SQL - NULL Values

- SQL uses the term **NULL** to represent a non-existent data value in the database. These values are not the same as an empty string or a zero. They don't hold any space in the database and are used to signify the absence of a value or the unknown value in a data field.

Example

Let us create a table with the name **CUSTOMERS** in the SQL database using the CREATE statement as shown in the query below –

```
</>
CREATE TABLE CUSTOMERS(
  ID INT NOT NULL,
  NAME VARCHAR (20) NOT NULL,
  AGE INT NOT NULL,
  ADDRESS CHAR (25) ,
  SALARY DECIMAL (18, 2),
  PRIMARY KEY (ID)
);
```

Let us insert some values into the above created table using the following query –

```
</>
INSERT INTO CUSTOMERS VALUES
(1, 'Ramesh', 32, 'Ahmedabad', 2000.00),
(2, 'Khilan', 25, 'Delhi', 1500.00),
(3, 'Kaushik', 23, 'Kota', 2000.00),
(4, 'Chaitali', 25, 'Mumbai', 6500.00),
(5, 'Hardik', 27, 'Bhopal', 8500.00),
(6, 'Komal', 22, 'Hyderabad', NULL),
(7, 'Muffy', 24, 'Indore', NULL);
```

The table is successfully created in the database.

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000.00
2	Khilan	25	Delhi	1500.00
3	Kaushik	23	Kota	2000.00
4	Chaitali	25	Mumbai	6500.00
5	Hardik	27	Bhopal	8500.00
6	Komal	22	Hyderabad	NULL
7	Muffy	24	Indore	NULL

Now, let us retrieve the records present in the table that are not null using the **IS NOT NULL** operator –

</>

Open Compiler

Edit & Run

```
SELECT ID, NAME, AGE, ADDRESS, SALARY
FROM CUSTOMERS
WHERE SALARY IS NOT NULL;
```

The above query would produce the following result –

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000.00
2	Khilan	25	Delhi	1500.00
3	Kaushik	23	Kota	2000.00
4	Chaitali	25	Mumbai	6500.00
5	Hardik	27	Bhopal	8500.00

You can also retrieve the NULL records present in the table using **IS NULL** operator in the SELECT query as shown below –

</>

Open Compiler

Edit & Run

```
SELECT ID, NAME, AGE, ADDRESS, SALARY
FROM CUSTOMERS
WHERE SALARY IS NULL;
```

The above query would produce the following result –

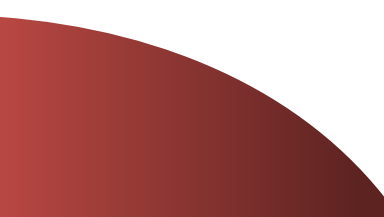
ID	NAME	AGE	ADDRESS	SALARY
6	Komal	22	Hyderabad	NULL
7	Muffy	24	Indore	NULL



Relational Algebra Operations

Division Operator

Section - 3.7



Division Operator

- ▶ Symbol: $\div$ (Division)
- ▶ Notation: $Relation1 (R1) \div Relation2 (R2)$ **OR** $Algebra1 \div Algebra2$
- ▶ Condition:
 - ➔ Attributes of relation2/algebra2 must be a proper subset of attributes of relation1/algebra1.
- ▶ Operation:
 - ➔ The output of the division operator will have attributes =
All attributes of relation1 – All attributes of relation2
 - ➔ The output of the division operator will have tuples =
Tuples in relation1, which are associated with the all tuples of relation2.

Division Operator Example

Example Perform Division operation between Student and Subject.

Student	
Name	Subject
Raj	DBMS
Raj	DS
Meet	DS
Meet	DF
Rohit	DBMS
Rohit	DS
Rohit	DF
Suresh	DBMS
Suresh	DF
Suresh	DS

Subject	
Subject	
DBMS	
DS	
DF	

Answer (Student) ÷ (Subject)

Output	
Name	
Rohit	
Suresh	

Division Operator Example

A	
Sno	PNo
S1	P1
S1	P2
S1	P3
S1	P4
S2	P1
S2	P2
S3	P2
S4	P2
S4	P4
S5	P4

B1	
PNo	
P2	

Algebra $(A) \div (B1)$

Output	
SNo	
S1	
S2	
S3	
S4	

B2	
PNo	
P2	
P4	

Algebra $(A) \div (B2)$

Output	
SNo	
S1	
S4	

B3	
PNo	
P1	
P2	
P4	

Algebra $(A) \div (B3)$

Output	
SNo	
S1	

B4	
PNo	
P2	
P5	

Algebra $(A) \div (B4)$

Output	
SNo	

Division Operator Example

Example List the **name of students** doing a **project in all technologies**.

Student		
RNo	Name	Technology
101	Raj	.NET
101	Raj	PHP
102	Meet	.NET
102	Meet	PHP
102	Meet	iPhone
102	Meet	Android
103	Rohit	Android
104	Suresh	.NET
104	Suresh	iPhone
104	Suresh	Android

Project	
TID	Technology
1	.NET
2	PHP
3	Android
4	iPhone

Answer $\Pi_{Name, Technology} (Student) \div \Pi_{Technology} (Project)$

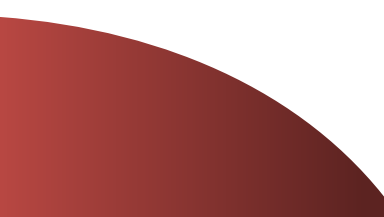
Output	
Name	
Meet	



Relational Algebra Operations

Rename Operator

Section - 3.8



Rename Operator

- ▶ Symbol: ρ (Rho)
- ▶ Notation: $\rho_{A(X1, X2, \dots, Xn)}$ (Relation)
- ▶ Operation:
 - The rename operation is used to **rename the output relation**.
 - The result of rename operator are also relations with new name.
 - The **original relation name can not be changed** when we perform rename operation on any relation.
- ▶ How to use:
 - $\rho_x(E)$
Returns a relation E under a new name X.
 - $\rho_{A1, A2, \dots, An}(E)$
Returns a relation E with the attributes renamed to A1, A2, ..., An.
 - $\rho_{x(A1, A2, \dots, An)}(E)$
Returns a relation E under a new name X with the attributes renamed to A1, A2, ..., An.

Rename Operator Example

Example Rename table

Student

RNo	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Algebra $\rho_{Person}(\text{Student})$

Person

RNo	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Example Rename attributes

Student

Rno	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Algebra $\rho_{(RollNo, StudentName, SPI)}(\text{Student})$

Student

RollNo	StudentName	SPI
101	Raj	8
102	Meet	9
103	Jay	7

Rename Operator Example

Example Rename table and attributes both

Student		
Rno	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Algebra $\rho_{Person (RollNo, StudentName)} (\pi_{RNo, Name} (Student))$

Person	
RollNo	StudentName
101	Raj
102	Meet
103	Jay

Example Rename particular attributes

Student		
Rno	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Algebra $\rho_{StudentName / Name} (Student)$

Student		
Rno	StudentName	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Rename Operator Example

Example Find out maximum CPI from student table.

Student		
Rno	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Step-2 $\sigma_{A.CPI < B.CPI} (\rho_A(\text{Student}) \times \rho_B(\text{Student}))$

Output-2					
A.Rno	A.Name	A.CPI	B.Rno	B.Name	B.CPI
101	Raj	8	102	Meet	9
103	Jay	7	101	Raj	8
103	Jay	7	102	Meet	9

Step-1 $\rho_A(\text{Student}) \times \rho_B(\text{Student})$

Output-1					
A.Rno	A.Name	A.CPI	B.Rno	B.Name	B.CPI
101	Raj	8	101	Raj	8
101	Raj	8	102	Meet	9
101	Raj	8	103	Jay	7
102	Meet	9	101	Raj	8
102	Meet	9	102	Meet	9
102	Meet	9	103	Jay	7
103	Jay	7	101	Raj	8
103	Jay	7	102	Meet	9
103	Jay	7	103	Jay	7

Rename Operator Example

Example Find out maximum CPI from student table.

Student		
Rno	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Step-3 $\Pi_{A.CPI} (\sigma_{A.CPI < B.CPI} (\rho_A(\text{Student}) \times \rho_B(\text{Student})))$

Output-3
A.CPI
8
7

Step-2 $\sigma_{A.CPI < B.CPI} (\rho_A(\text{Student}) \times \rho_B(\text{Student}))$

Output-2					
A.Rno	A.Name	A.CPI	B.Rno	B.Name	B.CPI
101	Raj	8	102	Meet	9
103	Jay	7	101	Raj	8
103	Jay	7	102	Meet	9

Rename Operator Example

Example Find out maximum CPI from student table.

Student		
Rno	Name	CPI
101	Raj	8
102	Meet	9
103	Jay	7

Step-3 $\Pi_{A.CPI} (\sigma_{A.CPI < B.CPI} (\rho_A(\text{Student}) \times \rho_B(\text{Student})))$

Output-3
A.CPI
8
7

Step-4 $\Pi_{CPI}(\text{Student}) - \Pi_{A.CPI} (\sigma_{A.CPI < B.CPI} (\rho_A(\text{Student}) \times \rho_B(\text{Student})))$

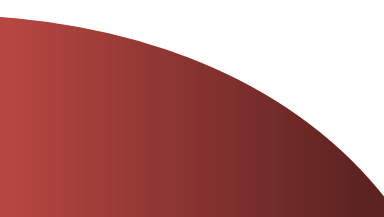
Student		Output-3		Output
CPI		A.CPI		CPI
8	-	8	=	9
9		7		
7				



Relational Algebra Operations

Aggregate Functions

Section - 3.9



Aggregate Functions

- ▶ Symbol: g or G
- ▶ Notation: $g_{function-name(column), function-name(column), ..., function-name(column)}(Relation)$
- ▶ Operation:
 - It **takes a more than one value** as input and **returns a single value** as output (result).
- ▶ Aggregate functions are:
 - Sum (It **returns the sum (addition)** of the values of a column.)
 - Max (It **returns the maximum** value for a column.)
 - Min (It **returns the minimum** value for a column.)
 - Avg (It **returns the average** of the values for a column.)
 - Count (It **returns total number** of values in a given column.)

Aggregate Functions Example

Student				
Rno	Name	Branch	Semester	CPI
101	Ramesh	CE	3	9
102	Mahesh	EC	3	8
103	Suresh	ME	4	7
104	Amit	EE	4	8
105	Anita	CE	4	8
106	Reeta	ME	3	7
107	Rohit	EE	4	9
108	Chetan	CE	3	8
109	Rakesh	CE	4	9

Example Find out sum of CPI of all students.

Answer $g_{sum(CPI)}(\text{Student})$

Output

sum

73

Example Find out maximum & minimum CPI.

Answer $g_{max(CPI), min(CPI)}(\text{Student})$

Output

max	min
-----	-----

9	7
---	---

Example Count the number of students.

Answer $g_{count(Rno)}(\text{Student})$

Output

count

9

Example Find out average of CPI of all students.

Answer $g_{avg(CPI)}(\text{Student})$

Output

avg

8.11

Relational Algebra [Exercise]

► Write down relational algebras for the following table:

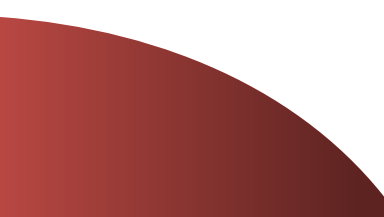
- Employee (person-name, street, city)
- Works (person-name, company-name, salary)
- Company (company-name, city)
- Managers (person-name, manager-name)

- ➔ Find the **names** of all employees who **work for “TCS”**.
- ➔ Find the **names** and **cities** of residence of all employees who **work for “Infosys”**.
- ➔ Find the **names**, **street** and **city** of residence of all employees **who work for “ITC” and earn more than \$10,000 per annum**.
- ➔ Find the **names** of all employees in this database who **live in the same city as the company for which they work**.
- ➔ Find the **names** of all employees **working in “TCS” who earn more than 25000 and less than 40000**.
- ➔ Find the **name** of employee **whose manager is “Ajay Patel” and salary is more than 50000**.
- ➔ Display the **name** of employee with **street, city, company name, salary and manager name staying in “Rajkot” and working in “Ahmedabad”**.
- ➔ Find **maximum, minimum and average salary** of all employee.
- ➔ Find out the **total number of employee**.



Open Source and Commercial DBMS

Section - 4



Open Source and Commercial DBMS

Open Source	Commercial DBMS
DBMS, which is available in the market at free of cost .	DBMS, which is available in the market at a certain price .
The code of open source DBMS product can be viewed, shared or modified by the community.	The code of commercial DBMS product can not be view, share or modify by the community.
There are chances of malfunctioning with code as source code is open.	The security is high and code is not accessible to unauthorized person.
Examples: MySQL, MongoDB, SQLite etc	Examples: Microsoft SQL Server, IBM Db2 etc

Questions asked in Exam

1. Define Super key, Primary key, Candidate key and Alternate key.
2. Explain following Relational Algebra Operation with example.
 - I. Selection
 - II. Projection
 - III. Cross Product
 - IV. Joins (Inner Join, Outer Joins)
 - V. Rename
 - VI. Division
 - VII. Set operators
3. Explain different aggregate functions with example.

Questions asked in Exam[Relational Algebra]

4. Consider the following relational database, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries

- employee (ssn, name, dno, salary, hobby, gender)
 - department (dno, dname, budget, location, mgrssn)
 - works_on (ssn, pno)
 - project (pno, pname, budget, location, goal)
- I. List all pairs of employee names and the project numbers they work on.
 - II. List out department number, department name and department budget.
 - III. List all projects that Raj Yadav works on by project name.
 - IV. List the names of employees who supervise themselves.

Questions asked in Exam [Relational Algebra]

5. Consider the following relational database, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries
- course (course-id, title, dept_name, credits)
 - instructor (id, name, dept_name, salary)
 - section (course-id, sec-id, semester, year, building, room_no, time_slot_id)
 - teaches (id, course-id, sec-id, semester, year)
- I. Find the name of all instructors in the physics department.
 - II. Find all the courses taught in the fall 2009 semester but not in Spring semester.
 - III. Find the names of all instructors in the Comp. Sci. department together with the course titles of all the courses that the instructors teach.
 - IV. Find the average salary in each department.

Questions asked in Exam[Relational Algebra]

6. Consider the following relations and write an relational algebra:

- EMP (empno, ename, jobtitle, managerno, hiredate, sal, commission, deptno)
- DEPT (deptno, dname, location)

- I. Find the Employees working in the department number 10, 20, 30 only.
- II. Find Employees whose names start with letter A or letter a.
- III. Find Employees along with their department name.
- IV. Find the Employees who are working in Smith's department
- V. Find the Employees who get salary more than Allen's salary.
- VI. Display employees who are getting maximum salary in each department.
- VII. Find list of employees whose hire date is on or before 1-April-18.

7. Consider the relational database given below and give an expression in the relational algebra:

- Employee (person-name, street, city) , Works (person-name, company-name, salary)
- Company (company-name, city) , Manages (person-name, manager-name)

- I. Find the names of all employees in this database who live in the same city as the company for which they work.
- II. Find the names, street address, and cities of residence of all employees who work for HCL and earn more than \$10,000 per annum.

Questions asked in Exam[Relational Algebra]

8. The relational database schema is given below and write the relational algebra expressions for the given queries.
- employee (person-name, street, city)
 - works (person-name, company-name, salary)
 - company (company-name, city)
 - manages (person-name, manager-name)
- I. Find the names of all employees who work for First Bank Corporation.
 - II. Find the names and cities of residence of all employees who work for First Bank Corporation.
 - III. Find the names, street address, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000 per annum.
 - IV. Find the names of all employees in this database who do not work for First Bank Corporation.